APE LEVEL / ENGINEERING DEVELOPMENT RECORD

RAGE REALMZ
WHISPER MARKET

Marketplace security, blockchain cancellation architecture, Oracle audit proofs, Water economy controls, and the live Water Well cooldown display.



Development session: November 2–3, 2026 | Environment: Polygon Amoy (chain ID 80002) | Prepared for: Ape Level

GitHub publication: Commit 9549df7, branch feature/polygon-wallet-auth. One untracked release-candidate ZIP remained intentionally excluded from the push.

1. Executive summary

The session began with unresolved Whisper Market listing and cancellation-security work and concluded with the successful activation of RAGE REALMZ's first server-controlled Water production economy. Development progressed through wallet authorization, cancellation receipt verification, Oracle signatures, automated recovery tests, production rate-limiting, historical inventory audits, and a live cooldown display inside the private Homestead.

The Water system now enforces a five-minute interval between collections and a maximum of 100 collected Water in a rolling 24-hour window, while retaining the existing 100-Water inventory capacity. The finished gameplay screenshot shows the well displaying "NEXT WATER: 04:34", replacing the indefinite "WELL CONSTRUCTED" prompt. The broader Water regression suite passed 242 tests with zero failures after fixture updates.

The blockchain-verifiable listing cancellation architecture was advanced substantially, but the cancellation contract remained undeployed and real wallet cancellation plus verified public WhisperLog publication were not enabled. These outstanding items are explicitly distinguished from completed functionality throughout this report.

Milestone	Confirmed outcome
Water production enforcement	Server-side SQLite transactional cooldown and rolling quota
Water and Homestead regression	242/242 passing after fixture updates
Countdown status tests	28/28 combined production/status tests
Client countdown compatibility	25/25 tests passing
Existing Water registry	35 registered; 34 available, 1 listed; 3 current owners
Cancellation architecture	Developed and tested components; not deployed or enabled
GitHub push	Commit 9549df7 on feature/polygon-wallet-auth

2. Initial Whisper Market issues

2.1 Player Water listing authorization

Work began around authenticated marketplace listings and wallet signatures. Coinbase Wallet EIP-712 structured signing required coordinated browser and server verification, including consistent domain data. Development addressed explicit EIP712Domain handling, wallet/provider context checks, and compatibility between submitted signatures and server validation. A player Water listing was ultimately created for 1 Water at 1.1 test TILT. This was an important end-to-end result for the listing pathway, not evidence that the wider trade and cancellation features were all complete.

2.2 Preventing unverified cancellation

The previous WebSocket cancellation pathway was blocked rather than left active while stronger authorization was unfinished. The intended replacement requires wallet-approved on-chain evidence on Polygon Amoy, verified separately against Whisper's ownership and listing records. A Solidity contract, WhisperListingCancellation, was created and compiled to record cancellation events. The contract verifies the transaction-level request, while Whisper must independently verify that the signing seller owns the specific active Water listing.

2.3 Transaction lifecycle and recovery

The cancellation request ledger tracks prepared, submitted, verifying, finalized, and rejected states, with replay and duplicate-submission protections. Receipt verification checks Amoy chain identity, transaction success, seller and contract addresses, calldata, event fields, confirmation count, and canonical block evidence. Recovery workers are designed to revisit interrupted requests without releasing Water twice or falsely treating an uncertain transaction as completed.

A key security rule remained enforced: missing confirmations, missing trusted publication, mismatched evidence, and simulated WhisperLog outages must defer or reject cancellation, leaving the Water listing and resource state protected.

3. Whisper Oracle audit integrity

The public Whisper repository's documented publication gate requires verification of the wallet, trade or operation, blockchain evidence, signed Oracle proof, successful publication, and read-back confirmation of the exact repository, proof contents, and commit SHA before final authorization. Failures or rate limits must not bypass the gate.

Cancellation-specific signed proofs were extended to include the TESTNET designation, explicit LISTING_CANCELLATION operation, and the unique Water resource ID. Tests verified that altering the transaction, seller, listing, block evidence, or resource identifier invalidates the proof. A separate Oracle authority test suite rejected untrusted or revoked Oracle keys. The combined signature/authority run passed 20/20 tests.

A later integration attempt to bind the verified Water ID through cancellation finalization failed one simulated recovery test because the mock acknowledgement was incomplete. That attempted change was automatically reverted. The publisher, secure key setup, durable public proof confirmation, final wallet flow,

and Amoy contract deployment remained outstanding.

4. Designing the Water economy

Ape Level adopted five economic principles for Whisper: accommodate the first player, cap resource creation, protect finite treasury budgets, separate player asking prices from treasury pricing, and set in-game resource prices without referencing TILT's external market price. Water was selected as the first fully specified production resource because Homestead Water Wells, inventory reservations, resource IDs, and recovery systems already existed.

Mechanic	Agreed policy
Water collection interval	One Water every 300 seconds
Hourly theoretical maximum	12 Water
Rolling daily production allowance	100 Water per 24 hours
Inventory holding capacity	100 Water
Well multiplier	Additional wells do not increase production
Offline accrual	No offline collection
Selling or transferring Water	Does not refund production allowance
Current identity enforcement	Per authenticated wallet; per verified human is future work

At five minutes per unit, a player can collect the full 100-Water daily allowance over approximately 8 hours and 20 minutes of active harvesting. Inventory capacity restricts holdings; the rolling production ledger independently restricts how much new Water a player can generate, even after selling previously harvested inventory.

5. Implementation: transactional production policy

5.1 Reusing the existing durable Water path

The audit located approximately 50 Water-related modules and tests. The authoritative function, collectDurableHomesteadWater(), synchronizes capacity from trusted Homestead state, generates a unique collection operation and Water ID, reserves SQLite capacity, records a durable Homestead event, and reconciles the two stores. Because SQLite and the filesystem cannot commit atomically together, interrupted operations require recovery rather than automatic capacity release.

5.2 Policy module and atomic reservation integration

A dedicated water-production-policy.mjs module defined a 300-second cooldown, 86,400-second rolling window, and 100-unit limit. Active reservation states (RESERVED, RECORDED, RECONCILED) count conservatively; RELEASED requests do not. New checks were integrated into reserveWaterProductionCapacity() inside the existing BEGIN IMMEDIATE SQLite transaction alongside inventory capacity checks. This guards against racing requests obtaining separate production permissions.

The standalone policy tests passed 13/13. The combined transactional enforcement, policy, and existing capacity suite passed 29/29. Tests covered immediate repeat attempts, the 299-second boundary, exactly 300 seconds, quota exhaustion, rollback on failure, duplicate operation IDs, and released reservation handling.

5.3 Backwards compatibility and recovery

Older regression tests intentionally harvested several Water immediately. The new cooldown correctly rejected those fixtures. Rather than weakening production enforcement, isolated test timestamps were spaced by five minutes. The focused recovery suite moved from 39/41 to 41/41. The broader Water/Homestead/inventory suite initially passed 240/242; after the buyer-production synchronization fixture was corrected, the entire 242/242 suite passed with no failures.

6. Persistent-state audit and activation safeguards

The live Whisper Market database was identified at private/whisper-market/market.sqlite, configured with SQLite write-ahead logging and synchronous FULL durability. The new production reservation table contained no historical reservations at the time of the audit. Separate inspection showed 35 existing registered Water units: 34 available, one listed, and three distinct current owners. These inventory records were not erased or reset.

A read-only scan of six Homestead JSON files found 50 WATER_MINT event occurrences, all older than the current rolling 24-hour window, with no invalid timestamps or unreadable files reported. Event occurrences should not be equated with distinct registered Water units. The evidence indicated that no recent-event backfill was required for initial activation.

Before restart, the team created a private backup under private/backups/water-economy-20261103-095551, containing a consistent SQLite database snapshot and the six Homestead JSON files. The snapshot passed SQLite integrity checking and reported a 270,336-byte database file. The backup remains local to the Codespace and is not an independent offsite recovery copy.

A controlled startup of the main game server on port 3000, with Polygon Amoy selected, was completed after resolving an occupied-port conflict. Live gameplay confirmed that immediate repeat Water collection was rejected with the message “Water Well is cooling down.”

7. Final feature: live Water Well cooldown display

The active restriction initially left an outdated interaction label: “WELL CONSTRUCTED.” That label communicated construction status rather than the next available harvest. Development therefore extended the server to calculate readiness from persistent reservation history and provide a server-time reference, next collection timestamp, remaining rolling allowance, and state (READY, COOLDOWN, or DAILY_LIMIT).

The client now displays a smoothly advancing countdown using elapsed browser time from the server-provided reference. Collection authorization remains exclusively server-side. The interface requests a refreshed readiness status after collection, and it clears stale status when a Homestead is re-entered. The status module’s combined suite passed 28/28, and the final client compatibility run passed 25/25.

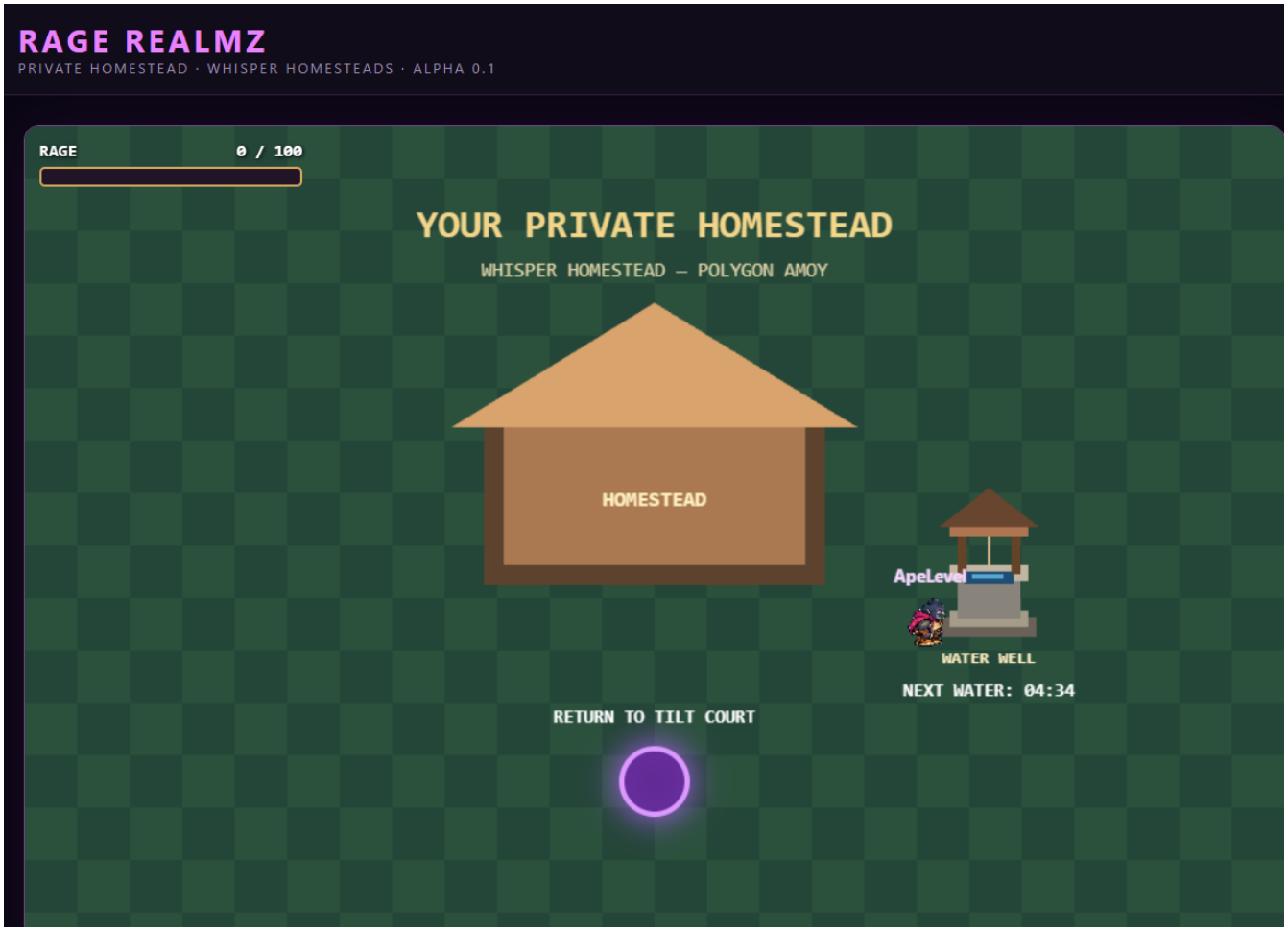


Figure 1. Live private Homestead gameplay proof showing “NEXT WATER: 04:34” directly beneath the constructed Water Well. The screenshot documents the successful interface change; server-side enforcement is supported separately by tests and the rejected immediate collection attempt.

The resulting experience replaces a static construction label with useful, in-world information. The screenshot is a visual success criterion, not independent proof that all economic or oracle controls are complete.

8. GitHub development publication

Following the development work, the user reported a successful GitHub commit and push for the RAGE REALMZ project. The reported commit identifier was 9549df7 on branch feature/polygon-wallet-auth. The reported push included Water economy and Homestead changes alongside broader Whisper Market work, smart contracts, cancellation security tests, game assets, NFT character animations, and release infrastructure. These figures are recorded from the supplied GitHub push summary; this report does not independently audit the commit diff.

Reported GitHub field	Value
Repository	ragequitterseth-lgtm/ragequitterz-game
Commit	9549df7
Branch	feature/polygon-wallet-auth
Files changed	299
Lines added	50,307
Lines removed	108
Data uploaded	100.13 MiB

Commit reference: https://github.com/ragequitterseth-lgtm/ragequitterz-game/commit/9549df7

Intentionally excluded: release-candidates/rage-realmz-seo-v0.1.1.zip remained an untracked local archive pending security review and a broader credential audit. Therefore the branch was reported synchronized for committed changes, not as containing every local file. No additional push of the excluded archive is recommended until review is complete.

9. Open work and engineering boundaries

- Listing cancellation: complete trustworthy WhisperLog publication and read-back verification; durable recovery of submissions interrupted before transaction-hash recording; configure the authorized Oracle signing identity; deploy the cancellation contract to Amoy through controlled authorization; connect wallet invocation and end-to-end tests.
- Economic model: define Rageberry growth, crafting recipes, potion consumption, treasury buying budgets, stock targets, pricing spreads, and anti-abuse controls for multiple accounts.
- Identity: move from temporary per-wallet limits to the planned per-verified-human production and inventory model, with linked-wallet recovery and abuse review.
- Operations: create a tested offsite backup and restore procedure; review the excluded release ZIP and scan for credentials before any additional publication; finish full release gating and security review before any

mainnet activation.

No Polygon mainnet contract deployment or token settlement is claimed as part of this Water cooldown milestone. Mainnet authorization remains subject to explicit approval.

10. Engineering conclusion

The session transformed Water harvesting from an unrestricted repeated interaction into a database-enforced production activity with clear scarcity rules, a rolling quota, recovery-aware reservations, and a visible in-world cooldown. The combination of automated testing, historical data checks, backups, activation and final gameplay screenshot supports the conclusion that the Water production and cooldown milestone was achieved.

At the same time, the cancellation and Oracle work should be described as a substantial security foundation rather than a deployed end-to-end product. Maintaining that distinction is important to the technical integrity of Ape Level’s public development record.

Final milestone: RAGE REALMZ displayed “NEXT WATER: 04:34” beneath the Water Well, with the five-minute production interval enforced by the Whisper server. The reported GitHub push recorded this development session under commit 9549df7.

END OF REPORT • APE LEVEL • NOVEMBER 3, 2026